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ONE LOVE. ONE FUTURE.

1

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Continuous Machine Learning

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2

2

CONTINUOUS MACHINE LEARNING

3

Classical Continuous Integration

- Previous sessions covered classical continuous integration from DevOps.
- Concepts applied to machine learning pipelines but could apply to any application.
- Now, we shift focus to **continuous machine learning** (CML).

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4

Continuous Machine Learning (CML)

- Focus on automating machine learning processes.
- Continuous integration principles are needed to ensure model reliability:
  - Did I train on the correct data?
  - Did my model converge?
  - Did it reach a threshold?
- CML allows continuous checking, with tools like `cml` from iterative.ai.

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5

Using CML in Workflows

- `cml` streamlines automation, providing reports after training.
- To trigger model training with every `git push`:

```
run: python train.py
```

- The workflow uses a standard CI process:
  - `push code` -> `trigger GitHub actions` -> `do stuff`.
  - A report is generated after each run.

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6

CML Workflow Overview

- The diagram shows the overall process using the `cml` framework.

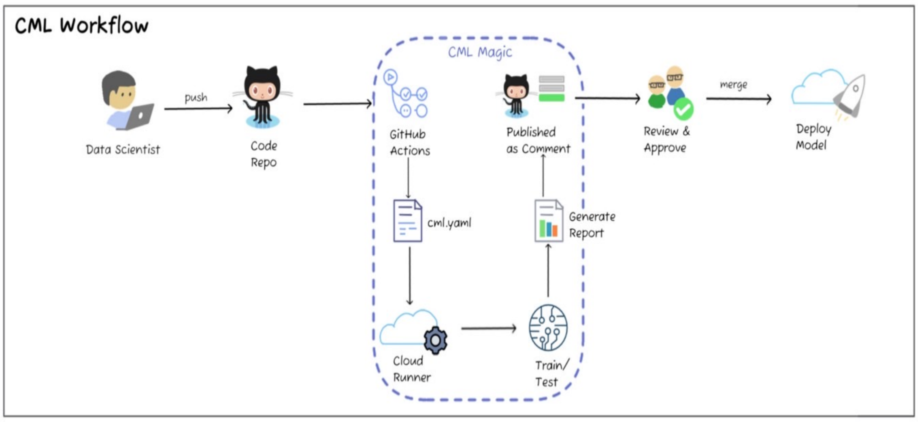

Image credit: towardsdatascience.com

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7

Exercises

1. Revisiting `train.py`

- Modify `train.py` to `train_cml.py` to report performance after training.
- Add accuracy and confusion matrix reporting:

```
# assume we have a trained model
import matplotlib.pyplot as plt
from sklearn.metrics import classification_report, confusion_matrix,
ConfusionMatrixDisplay
preds, target = [], []
for batch in train_dataloader:
    x, y = batch
    probs = model(x)
    preds.append(probs.argmax(dim=-1))
    target.append(y.detach())

target = torch.cat(target, dim=0)
preds = torch.cat(preds, dim=0)

report = classification_report(target, preds)
with open("classification_report.txt", 'w') as outfile:
    outfile.write(report)
confmat = confusion_matrix(target, preds)
disp = ConfusionMatrixDisplay(confusion_matrix = confmat)
plt.savefig('confusion_matrix.png')
```

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8

2. Create a CML Workflow

- Create a new workflow file `cml.yaml` with the following content:

```
name: train-my-model
on: [push]
jobs:
  run:
    runs-on: [ubuntu-latest]
    steps:
      - uses: actions/checkout@v2
      - uses: iterative/setup-cml@v1
      - name: Train model
        run: |
          pip install -r requirements.txt  # install dependencies
          python train.py  # run training
      - name: Write report
        env:
          REPO_TOKEN: ${{ secrets.GITHUB_TOKEN }}
        run: |
          cat classification_report.txt >> report.md
          cml-publish confusion_matrix.png --md >> report.md
          cml-send-comment report.md
```

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9

3. Push Workflow File

- Push `cml.yaml` to GitHub and ensure the workflow completes.

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10

4. Send a Pull-Request

- Watch this video on how to send a pull-request.
- Ensure the workflow generates a performance report in a GitHub comment.

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5. (Optional) Deep Dive into `cml` and `dvc`

- Learn how `cml` integrates with `dvc` here.

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12

CML and Hardware Limitations

- GitHub Actions default runners are CPU-only machines.
- Modern machine learning often requires GPUs for reasonable training times.
- After learning about cloud computing, experiment with self-hosted runners.

Continual Learning

Model's performance degrades in production

- Data distribution shifts
  - Sudden
  - Cyclic
  - Gradual

15

From Monitoring to Continual Learning

- Monitoring: detect changing data distributions
- Continual learning: continually adapt models to changing data distributions

16

Continual learning

- Set up infrastructure such that models can continuously learn from new data in production
- Stateful training

17

Continual learning: use cases

- Rare events
  - Christmas/Black Friday/Prime Day shopping
  - Total Landscaping
- Continuous cold start (in-session adaptation)
  - New users
  - New devices
  - Users not logged in
  - Users rarely logged in

18

Continual learning is especially good for

- Natural labels: e.g. user click -> good prediction
- Short feedback loops
- Examples:
  - RecSys
  - Ranking
  - Ads CTR prediction
  - eDiscovery

19

How often to retrain your model is just a knob to turn

20

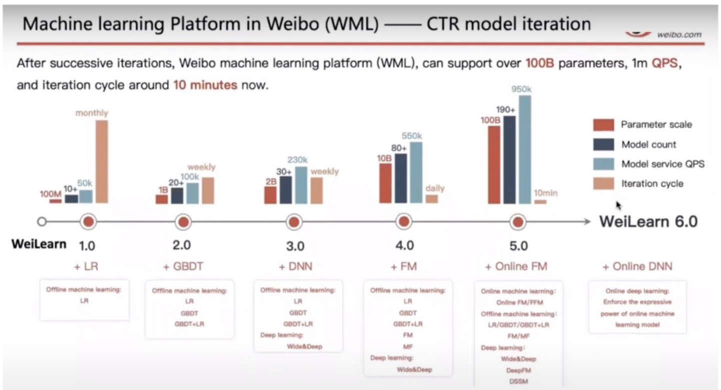
How frequently should you update your models?

- Very few companies actually update models with each incoming sample
  - Catastrophic forgetting
  - Can get unnecessarily expensive*
- Update models with micro-batches

21

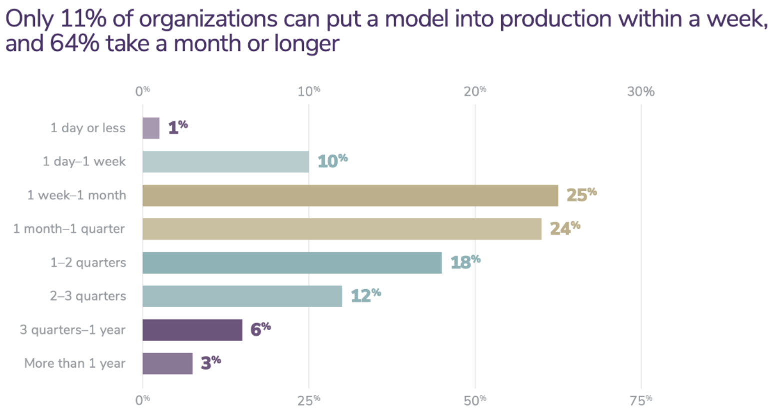
Iteration cycle: minutes

- Alibaba: Singles Day sale
- Weibo
- Tiktok
- SheIN



22

Iteration cycle: US



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Algorithmia's State of Enterprise Machine Learning (2021)

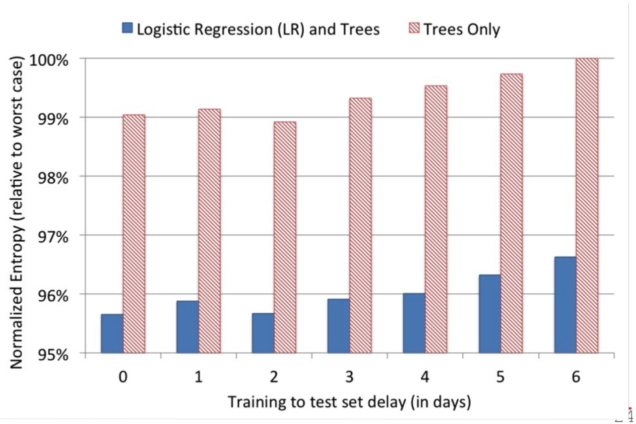
23

23

Quantify the value of data freshness

1. How much model's performance changes if switch from retraining monthly to weekly to daily to hourly?
   a. FB: CTR loss can be reduced ~1%

      going from training weekly to daily



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Practical Lessons from Predicting Clicks on Ads at Facebook (2014)

24

Quantify the value of data freshness

1. How much model's performance changes if switch from retraining monthly to weekly to daily to hourly?
2. How would retention change if you can do in-session adaptation?

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25

25

Quantify the value of data freshness

1. How much model's performance changes if switch from retraining monthly to weekly to daily to hourly?
2. How would retention change if you can do in-session adaptation?
3. Model iteration vs. data iteration

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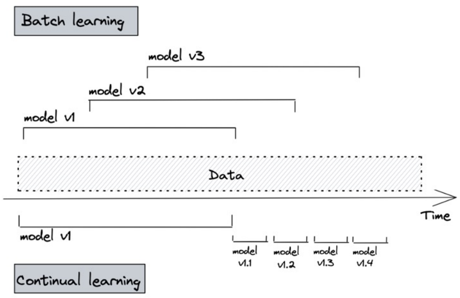
26

26

Quantify cloud bill savings

- Train model incrementally each day on fresh data
- Faster convergence → less compute needed



Going fr ... ng gives

GRUBHUB™

45x cost savings and **+20% metrics increase**

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Online Learning for Recommendations at Grubhub (Alex Egg, 2021)

27

27

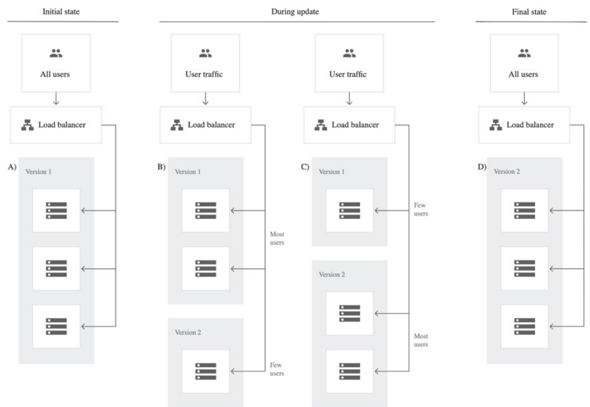
Test in Production

30

30

Canary testing

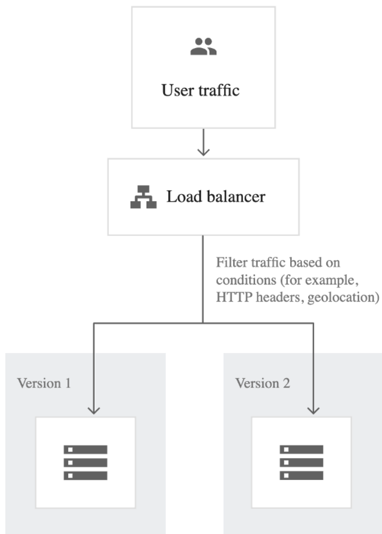
- New model alongside existing system
- Some traffic is routed to new model
- Slowly increase the traffic to new model
  - E.g. roll out to Vietnam first, then Asia, then rest of the world



Application deployment and testing strategies (Google)

A/B testing

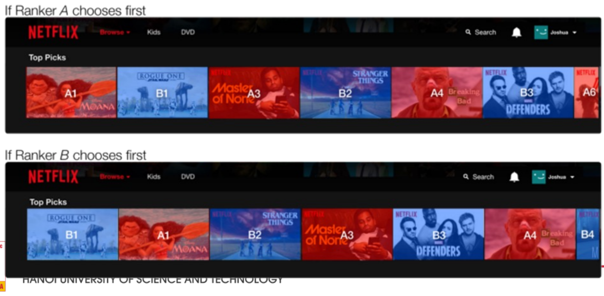
- New model alongside existing system
- A percentage of traffic is routed to new model based on routing rules
- Control target audience & monitor any statistically significant differences in user behavior
- Can have more than 2 versions

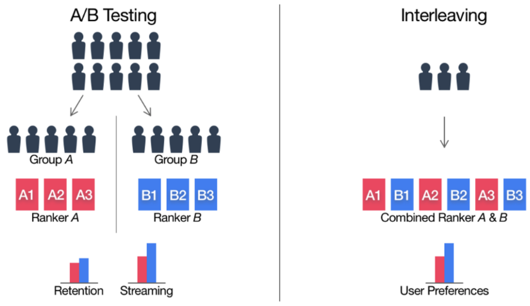


Application deployment and testing strategies (Google)

Interleaved experiments

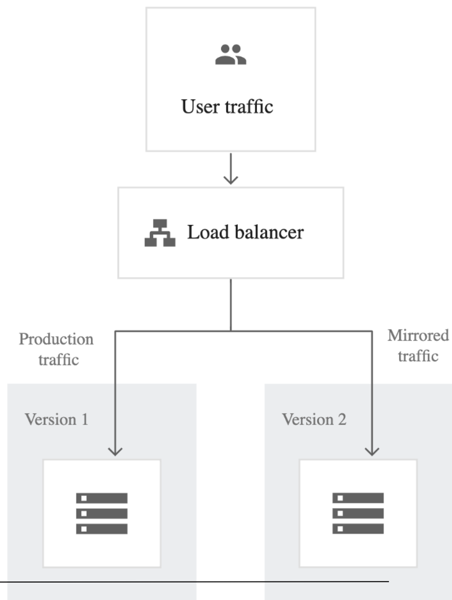
- Especially useful for ranking/recsys
- Take recommendations from both model A & B
- Mix them together and show them to users
- See which recommendations are clicked on





Shadow testing

- New model in parallel with existing system
- New model's predictions are logged, but not show to users
- Switch to new model when results are satisfactory



[Application deployment and testing strategies](#) (Google)

Test internally first

- Use features even as they're in development
- Share internally before externally

⚠️ You and your coworkers are not typical users ⚠️

35

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36